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AI INTERACTION TRANSCRIPT

MA677 — Fisher Information and the Cramér-Rao Inequality

Tool: Claude (Anthropic, claude.ai)

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PURPOSE

I used Claude as a study aid to work through the mathematics in Cover & Thomas, Elements of Information Theory (2nd ed.), Section 11.10: Fisher Information and the Cramér-Rao Inequality.

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SESSION 1 — Understanding the Score Function

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ME:

I am reading Cover & Thomas Section 11.10. Can you explain what the score function is and why it matters?

CLAUDE:

The score function V is defined as the derivative of the log-likelihood with respect to the parameter θ:

$$V = \partial/\partial\theta \ln f(X; \theta)$$

Intuitively, it measures how sensitive the log-likelihood is to small changes in θ. If the score is large in magnitude, the data reacts strongly when θ shifts — meaning the data is informative about θ. If the score is small, the distribution is flat and the data is less useful for pinning down θ.

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SESSION 2 — Why Does EV = 0?

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ME:

I don't understand why the expected value of the score is always zero. The book just states it. Can you walk me through it?

CLAUDE:

Sure. The key is that f(x; θ) is a proper probability density, so it always integrates to 1:

$$\int f(x; \theta)\, dx = 1 \quad \text{for all } \theta$$

Now differentiate both sides with respect to θ:

$$\frac{\partial}{\partial\theta} \int f(x;\theta)\,dx = 0$$

Swapping the derivative and integral (valid for well-behaved densities by the bounded convergence theorem):

$$\int \frac{\partial}{\partial\theta} f(x;\theta)\,dx = 0$$

Now apply the log-derivative trick. By the chain rule:

$$\frac{\partial}{\partial\theta} f(x;\theta) = f(x;\theta) \cdot \frac{\partial}{\partial\theta} \ln f(x;\theta)$$

Substituting back:

$$\int \left[\frac{\partial}{\partial\theta} \ln f(x;\theta)\right] \cdot f(x;\theta)\,dx = 0$$

But that integral is exactly $E[V]$ — the expected value of the score. So $EV = 0$.

ME:
Why does the log-derivative trick work?

CLAUDE:
It is just the chain rule applied to the natural log. If you differentiate $\ln(g)$ with respect to θ, you get g'/g. So:

$$\frac{\partial}{\partial\theta} \ln f(x;\theta) = \left[\frac{\partial}{\partial\theta} f(x;\theta)\right] / f(x;\theta)$$

Rearranging:

$$\frac{\partial}{\partial\theta} f(x;\theta) = f(x;\theta) \cdot \frac{\partial}{\partial\theta} \ln f(x;\theta)$$

This identity is useful because it lets you convert derivatives of f into expectations involving the score, which is algebraically much cleaner.

ME:
So why does $EV = 0$ make intuitive sense?

CLAUDE:
Because probabilities must always sum to 1. As θ changes, the distribution shifts — some outcomes become more likely, others less likely. But the total probability cannot change. The score tracks how the density redistributes itself as θ moves, and because the total must stay fixed at 1, the average redistribution must be zero. The score cannot consistently push in one direction.

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SESSION 3 — Fisher Information
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ME:
Since $EV = 0$, the variance of V is just $E[V^2]$. Is that how Fisher information is defined?

CLAUDE:
Exactly. Because $EV = 0$, we have:

$$\mathrm{Var}(V) = E[V^2] - (EV)^2 = E[V^2]$$

And Fisher information is defined as this variance:

$$J(\theta) = E_\theta\left[\left(\partial/\partial\theta \ln f(X; \theta)\right)^2 \right]$$

It measures the average squared sensitivity of the log-likelihood to θ. High $J(\theta)$ means each observation carries a lot of information about θ. Low $J(\theta)$ means observations are nearly uninformative.

ME:
What happens with n i.i.d. observations?

CLAUDE:
For a sample $X_1, \ldots, X_n$ drawn i.i.d. from $f(x; \theta)$, the joint density is the product of the individual densities. The score of the joint density is the sum of the individual scores:

$$V(X_1,\ldots,X_n) = \sum_{i=1}^{n} V(X_i)$$

Since the $V(X_i)$ are i.i.d. with mean 0 and variance $J(\theta)$, the variance of their sum is:

$$J_n(\theta) = n \cdot J(\theta)$$

So the Fisher information scales linearly with sample size — more data means more information, as expected.

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SESSION 4 — The Cramér-Rao Inequality
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ME:
Can you walk me through the proof of the Cramér-Rao bound?

CLAUDE:
The proof uses the Cauchy-Schwarz inequality. Let V be the score

and T be any unbiased estimator of θ. Cauchy-Schwarz says:

$$(E[(V - EV)(T - E_\theta T)])^2 \leq E[(V - EV)^2] \cdot E[(T - E_\theta T)^2]$$

Since T is unbiased, $E_\theta T = \theta$. Since $EV = 0$, this simplifies to:

$$(E[VT])^2 \leq \mathrm{Var}(V) \cdot \mathrm{Var}(T)$$
$$(E[VT])^2 \leq J(\theta) \cdot \mathrm{Var}(T)$$

Now we need to compute $E[VT]$. By definition of V:

$$\begin{aligned} E[VT] &= \int [\partial/\partial\theta \ln f(x;\theta)]\, T(x)\, f(x;\theta)\, dx \\ &= \int [\partial/\partial\theta f(x;\theta)]\, T(x)\, dx \\ &= \partial/\partial\theta \int f(x;\theta)\, T(x)\, dx \\ &= \partial/\partial\theta\, E_\theta[T] \\ &= \partial/\partial\theta\, \theta \\ &= 1 \end{aligned}$$

So $E[VT] = 1$. Substituting back into the Cauchy-Schwarz result:

$$1 \leq J(\theta) \cdot \mathrm{Var}(T)$$

Dividing both sides by $J(\theta)$:

$$\mathrm{Var}(T) \geq 1/J(\theta)$$

This is the Cramér-Rao inequality.

ME:
What does it mean for an estimator to be efficient?

CLAUDE:
An unbiased estimator T is called efficient if it achieves the Cramér-Rao lower bound with equality:

$$\mathrm{Var}(T) = 1/J(\theta)$$

This means no other unbiased estimator can have smaller variance. The efficient estimator extracts the maximum possible information from the data.

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SESSION 5 — Worked Example: Bernoulli Distribution
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ME:
Can you work through the Fisher information for Bernoulli(θ)?

CLAUDE:
Sure. For $X \sim \text{Bernoulli}(\theta)$:

$$f(x;\theta) = \theta^x (1-\theta)^{(1-x)}, \quad x \in \{0,1\}$$

Taking the log:

$$\ln f(x;\theta) = x \ln \theta + (1-x) \ln(1-\theta)$$

The score function is:

$$V = \partial/\partial\theta \ln f(X;\theta) = X/\theta - (1-X)/(1-\theta)$$

Verify $EV = 0$:

$$\begin{aligned} E[V] &= E[X]/\theta - (1 - E[X])/(1-\theta) \\ &= \theta/\theta - (1-\theta)/(1-\theta) \\ &= 1 - 1 = 0 \quad ✓ \end{aligned}$$

Fisher information:

$$J(\theta) = E[V^2] = E[(X/\theta - (1-X)/(1-\theta))^2]$$

Expanding and using $E[X] = \theta$, $E[X^2] = \theta$ (since $X \in \{0,1\}$):

$$J(\theta) = 1/[\theta(1-\theta)]$$

The Cramér-Rao bound for n observations is:

$$\text{Var}(T) \geq \theta(1-\theta)/n$$

The sample proportion p-hat = $(X1+...+Xn)/n$ has:

$$\text{Var}(\text{p-hat}) = \theta(1-\theta)/n = 1/J_n(\theta)$$

So p-hat is efficient — it achieves the bound exactly.

ME:
Why is $J(\theta)$ largest near $\theta = 0$ and $\theta = 1$?

CLAUDE:
Because $J(\theta) = 1/[\theta(1-\theta)]$, which blows up as θ approaches 0 or 1. Intuitively, when θ is close to 0 or 1, the distribution is highly skewed — nearly every observation is the same value. Each observation is therefore very informative: seeing a 1 when θ is near 0 is a strong signal. By contrast, when $\theta = 0.5$, the distribution is maximally uncertain — each observation is a fair coin flip and tells you relatively little about θ.

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END OF TRANSCRIPT
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